

Clay Ndugga

SOFTWARE ENGINEER

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Education

Applied Mathematics and Engineering

Kingston, ON

QUEEN'S UNIVERSITY

2024

- Similar to **Software Engineering** with a focus on advanced mathematics
- **Relevant Coursework:** Statistical Modelling, Data structure's and Algorithms, Stochastic Processes, Operating Systems, Embedded Systems, Control Theory, Information Theory, Optimization and Control of Stochastic Systems, Data Compression, Computer Networks

AWS Certified Solutions Architect (SAA-CO3)

2024 Certification

Work Experience

Software Engineer Intern

Calgary, AB

LONGVIEW SYSTEMS

May 2023–Sep 2023

- Built a **stateful REST API data pipeline** (ETL) with queuing to ingest **1M+** employee records, reducing ingestion times by 70%. Implemented idempotent retries, pagination tracking, and checkpointing to handle API rate limits, enabling 24/7 reliability on spot instances. Later adopted as the team's standard ingestion template.
- Migrated 20k+ lines of legacy SQL stored procedures to move a to modern **Databricks/Pyspark** architecture drastically cutting run times.
- Automated Azure wiki documentation for 300+ functions using **Open AI API** (LLM's) eliminating 15+ hours/work of manual effort. Designed a structured prompting process to reduce hallucinations achieving 100% accuracy.
- **Technologies:** Azure, Databricks, Python, PySpark, SQL

Machine Learning Engineer Intern

Calgary, AB

BOARDWALK REAL ESTATE

May 2022 – Sept 2022

- Built an **ML model** using **Tensorflow** to predict tenant lease renewals (**binary classification**) achieving a 0.85 (out of 1) **F1-score**, reducing profit loss to empty properties
- Created **data pipelines** to ingest and preprocess **1M+ records** from SQL databases, surveys, and property reports using Pandas and SQL
- Optimized model precision via **hyperparameter tuning** (fine tuning) to prioritize the reduction of false positives to minimize vacancy risk (precision recall)
- Periodically communicated technical concepts to non-technical stakeholders (The Board of Directors)
- **Technologies:** Python, SQL, Jupyter, Pandas, NumPy, Scikit-Learn, TensorFlow

Software Engineer Intern

Calgary, AB

LONGVIEW SYSTEMS

May 2021 – Sept 2021

- Architected an **end-to-end IoT Azure cloud solution** to optimize building energy efficiency using **real time machine learning** for decision making
- Created robust data pipeline to stream in **real-time unstructured data** from IOT temperature sensors. Scheduled jobs to handle the process of cleaning and aggregating data for future ML modeling
- Used **Random Forest Regression** for **times series forecasting** to determine optimal setpoints 24h into the future.
- Employed best practice ML Ops by **automating monthly retraining** via **MLflow** to adapt to seasonal patterns, increasing forecasting accuracy.
- **Technologies:** Azure, Databricks, Python, MLOps, MLFlow, SQL

Thesis

Deep Learning Point Cloud Compression via Non-Linear Transform Coding

[🔗 Link](#)

3D COMPUTER VISION RESEARCH

- Designed a hybrid **FoldingNet encoder/PCN decoder** architecture achieving high compression ratios on the ShapeNet dataset, outperforming voxel-based methods in rate-distortion trade-offs.
- Engineered a **quantization-aware training pipeline** in TensorFlow, implementing Gaussian entropy modeling for latent space compression, reducing bitrate by 22% while maintaining reconstruction quality.
- Optimized point cloud batch processing using **K-NN graph layers**, enabling training on 13,000+ models (2048 pts each) reducing GPU memory reduction vs. voxel approaches.
- **Technologies:** Python, TensorFlow, LiDAR, Chamfer Distance, Rate-Distortion Optimization

Projects

Chat with PDF (RAG)

 [Link](#)

WEB APPLICATION

- Designed and deployed a scalable **RAG-based PDF chatbot** for multi document Q&A with clickable references to the source material
- Built **automated evaluation pipeline**: Created a test suite of 100+ query/response pairs, benchmarking prompt iterations via cosine similarity against human-annotated ground truths, cutting manual review time
- Optimized search relevance by integrating cross-encoder **reranking** (Pinecone) atop **vector search**, improving answer precision for multi-document queries with conflicting data.
- **Technologies**: *Python, OpenAI, Pinecone, Docker, Google Cloud (Cloud Run), Node.js, Typescript, React, Firebase*

YouTube Music Discovery Engine

 [Link](#)

SERVERLESS WEB APPLICATION

- Architected a fault-tolerant **AWS web application** to identify songs from Youtube comments. Combined **SQS queues, DynamoDB** job tracking, and 15+ Lambda functions to achieve **scalability** and **performance**
- Designed q-gram/Jaccard **similarity algorithm** for song identification to reduced duplicate Spotify API calls while maintaining match accuracy across the comment dataset
- Implemented idempotent job processing with hierarchical DynamoDB entries, preventing redundant compute costs through channel/video job de duplication
- Built S3 **cache** for YouTube/Spotify data cutting comment processing time with TTL-based invalidation
- **Technologies**: *AWS (Lambda, SQS, DynamoDB, S3), Next.js, TypeScript, Python, React*

Skills

ML Topics: Tensorflow | Pytorch | Scikit-learn | Pandas | XGBoost | Computer Vision | NLP | RAG

Dev-ops & Infrastructure: Azure | Databricks | AWS | Docker | Git | Linux

Other: Python | SQL | MATLAB | Javascript | Typescript | C | HTML/CSS | React | NodsJS | ExpressJS